

"Start Symbol" = <PROGRAM>

! ----- Terminals ----- ! Atualizado em 16/09/2023 às 22:03

Decimal = {Number}+<{'.' {Number}+}?>

IDE = {Letter} ({Letter} | {Number} | '_')*

{String Chars} = {Printable} - {NBSP} - ["]

StringLiteral = "" {String Chars}* ""

Bool = 'true' | 'false'

ART = '+' | '-' | '*' | '/'

ART_DOUBLE = '++' | '--'

REL = '<' | '<=' | '>' | '>=' | '!=' | '=='

LOG = '&&' | '||'

! ----- Rules-----

! Atualizado em 20/09/2023 às 20:16

<DEC_OBJECT_ATTRIBUTE_ACCESS> ::= IDE <DIMENSIONS>

<END_OBJECT_ATTRIBUTE_ACCESS>

! Atualizado em 09/09/2023 às 16:13

<NRO> ::= Decimal

! Atualizado em 09/09/2023 às 16:13

<CAC> ::= StringLiteral

! Atualizado em 14/09/2023 às 20:52

<TYPE> ::= 'int'

| 'real'

| 'boolean'

| 'string'

! Atualizado em 14/09/2023 às 20:52

<DIMENSIONS> ::= '[' <SIZE_DIMENSION> ']' <DIMENSIONS>

|

! Atualizado em 14/09/2023 às 20:52

<SIZE_DIMENSION> ::= IDE

| <NRO>

! Atualizado em 14/09/2023 às 20:59

<DEC_VAR> ::= IDE <DIMENSIONS>

! Atualizado em 16/09/2023 às 21:50

<PARAMETERS> ::= <VALUE> <MULT_PARAMETERS>

|

! Atualizado em 16/09/2023 às 21:50

<MULT_PARAMETERS> ::= ',' <VALUE> <MULT_PARAMETERS>

|

<OBJECT_METHOD_OR_OBJECT_ACCESS_OR_PART> ::=

<DEC_OBJECT_ATTRIBUTE_ACCESS> <OPTIONAL_OBJECT_METHOD_ACCESS>

! -----

! Bloco declaração parâmetros construtor - Atualizado em 18/09/2023 às 19:15

<DEC_PARAMETERS_CONSTRUCTOR> ::= <MULT_PARAM_CONSTRUCTOR>

<MULT_DEC_PARAMETERS_CONSTRUCTOR>

|

<VARIABLE_PARAM> ::= <TYPE> IDE

<OBJECT_PARAM> ::= IDE IDE

<MULT_DEC_PARAMETERS_CONSTRUCTOR> ::= ',' <MULT_PARAM_CONSTRUCTOR>

<MULT_DEC_PARAMETERS_CONSTRUCTOR>

|

<MULT_PARAM_CONSTRUCTOR> ::= <VARIABLE_PARAM>

| <OBJECT_PARAM>

! -----

! Bloco de variáveis - Atualizado em 14/09/2023 às 20:38

<VARIABLES_BLOCK> ::= 'variables' '{' <VARIABLES>

<VARIABLES> ::= <VARIABLE> <VARIABLES>

| '}'

<VARIABLE> ::= <TYPE> <DEC_VAR> <MULTIPLE_VARIABLES_LINE>

<MULTIPLE_VARIABLES_LINE> ::= ','

| ',' <DEC_VAR> <MULTIPLE_VARIABLES_LINE>

! -----

! Bloco de objetos - Atualizado em 14/09/2023 às 20:59

<OBJECTS_BLOCK> ::= 'objects' '{' <OBJECTS>

<OBJECTS> ::= <OBJECT> <OBJECTS>

| '}'

<OBJECT> ::= IDE <DEC_VAR> <MULTIPLE_OBJECTS>

<MULTIPLE_OBJECTS> ::= ',' <DEC_VAR> <MULTIPLE_OBJECTS>
| ','

! -----

! Bloco de classes - Atualizado em 18/09/2023 às 17:19

<CLASS_BLOCK> ::= 'class' <IDE_CLASS>

<IDE_CLASS> ::= IDE <EXTENDS>
| <MAIN>

<EXTENDS> ::= 'extends' IDE <START_CLASS_BLOCK>
| <START_CLASS_BLOCK>

<START_CLASS_BLOCK> ::= '{' <INIT_CLASS>

<INIT_CLASS> ::= <BODY_BLOCKS> <METHODS_BLOCK> <CONSTRUCTOR>

<CONSTRUCTOR> ::= 'constructor' '(' <DEC_PARAMETERS_CONSTRUCTOR> ')' '{'
<VARIABLES_BLOCK> <OBJECTS_BLOCK> <COMMANDS> '}' <END_CLASS>

<END_CLASS> ::= '}' <CLASS_BLOCK>

<MAIN> ::= 'main' '{' <INIT_MAIN>

<INIT_MAIN> ::= <BODY_BLOCKS> <MAIN_METHODS> '}'

<BODY_BLOCKS> ::= <VARIABLES_BLOCK> <OBJECTS_BLOCK>

! -----

! Bloco da main - Atualizado em 18/09/2023 às 17:16

<MAIN_METHODS> ::= 'methods' '{' <MAIN_METHODS_BODY> '}'

<MAIN_METHODS_BODY> ::= <MAIN_TYPE> 'main()' '{' <METHOD_BODY>
<METHODS>

<MAIN_TYPE> ::= <TYPE>
| 'void'

! -----

! Bloco de if-else - Atualizado em 14/09/2023 às 08:30

<IF> ::= 'if' '(' <CONDITION> ')' 'then' '{' <COMMANDS> '}' <IF_ELSE>

<IF_ELSE> ::= 'else' '{' <COMMANDS> '}'
|

<CONDITION> ::= <LOGICAL_EXPRESSION>

!-----

! Bloco de constantes - Atualizado em 14/09/2023 às 20:49

<CONSTS_BLOCK> ::= 'const' '{' <CONSTS>

<CONSTS> ::= <CONST> <CONSTS>
| '}'

<CONST> ::= <TYPE> <CONST_ATRIBUTION> <MULTIPLE_CONSTS>

<CONST_ATRIBUTION> ::= IDE '=' <ATRIBUTION>

<MULTIPLE_CONSTS> ::= ',' <CONST_ATRIBUTION> <MULTIPLE_CONSTS>
| ';'

<ATRIBUTION> ::= <NRO>
| Bool
| <CAC>

!-----

! Bloco atribuição de valor - Atualizado em 19/09/2023 às 19:53

<OBJECT_ACCESS_OR_ASSIGNMENT> ::= <DEC_OBJECT_ATTRIBUTE_ACCESS>
<OBJECT_ACCESS_OR_ASSIGNMENT_END>

<OBJECT_ACCESS_OR_ASSIGNMENT_END> ::= <OBJECT_METHOD_ACCESS_END> |
'=' <VALUE> | ART_DOUBLE

<VALUE> ::= <NRO> <SIMPLE_OR_DOUBLE_ARITHMETIC_EXPRESSION_OPTIONAL>
| <CAC>
| <VECTOR_ASSIGN_BLOCK>
| <INIT_EXPRESSION>
| '!' <LOGICAL_EXPRESSION_BEGIN> <LOGICAL_EXPRESSION_END>
| <ARITHMETIC_OR_LOGICAL_EXPRESSION_WITH_PARENTHESES>
| Bool

<SIMPLE_OR_DOUBLE_ARITHMETIC_EXPRESSION_OPTIONAL> ::=
<SIMPLE_OR_DOUBLE_ARITHMETIC_EXPRESSION>
|

<INIT_EXPRESSION> ::= <DEC_OBJECT_ATTRIBUTE_ACCESS>
<ARITHMETIC_OR_LOGICAL_EXPRESSION>

<ARITHMETIC_OR_LOGICAL_EXPRESSION> ::=
<OPTIONAL_OBJECT_METHOD_ACCESS> <LOG_REL_OPTIONAL>
<LOGICAL_EXPRESSION_END>
| <SIMPLE_OR_DOUBLE_ARITHMETIC_EXPRESSION>

<ARITHIMETIC_OR_LOGICAL_EXPRESSION_WITH_PARENTHESES> ::=
<PARENTHESES_BEGIN>

<PARENTHESES_BEGIN> ::= '(' <EXPRESSIONS> <PARENTHESES_END>

<EXPRESSIONS> ::= <PARENTHESES_BEGIN>
| <SIMPLE_EXPRESSION_WITHOUT_PARENTHESES>
| <LOGICAL_EXPRESSION_WITHOUT_PARENTHESES>
| <SIMPLE_OR_LOGICAL_IDE_BEGIN>

<SIMPLE_OR_LOGICAL_IDE_BEGIN> ::= <DEC_OBJECT_ATTRIBUTE_ACCESS>
<SIMPLE_OR_LOGICAL_IDE_END>

<SIMPLE_OR_LOGICAL_IDE_END> ::= <END_EXPRESSION> |
<OPTIONAL_OBJECT_METHOD_ACCESS> <LOG_REL_OPTIONAL>
<LOGICAL_EXPRESSION_END>

<PARENTHESES_END> ::= ')' <EXPRESSIONS_WITHOUT_PARENTHESES_END>

<EXPRESSIONS_WITHOUT_PARENTHESES_END> ::= <END_EXPRESSION>
| LOG <LOGICAL_EXPRESSION_BEGIN> <LOGICAL_EXPRESSION_END>
|

<SIMPLE_EXPRESSION_WITHOUT_PARENTHESES> ::= <NRO> <END_EXPRESSION>

<LOGICAL_EXPRESSION_WITHOUT_PARENTHESES> ::= Bool
<LOGICAL_EXPRESSION_END>
| '!' <LOGICAL_EXPRESSION_BEGIN> <LOGICAL_EXPRESSION_END>

<VECTOR_ASSIGN_BLOCK> ::= '[' <ELEMENTS_ASSIGN> ']'

<ELEMENTS_ASSIGN> ::= <ELEMENT_ASSIGN> <MULTIPLE_ELEMENTS_ASSIGN>

<N_DIMENSIONS_ASSIGN> ::= '[' <ELEMENTS_ASSIGN> ']'
|

<MULTIPLE_ELEMENTS_ASSIGN> ::= ',' <ELEMENT_ASSIGN>
<MULTIPLE_ELEMENTS_ASSIGN>
|

<ELEMENT_ASSIGN> ::= IDE
| StringLiteral
| <NRO>
| <N_DIMENSIONS_ASSIGN>

!-----

! Bloco print e read - Atualizado em 14/09/2023 às 22:25

<PRINT_BEGIN> ::= 'print' '(' <PRINT_END>

<PRINT_END> ::= <PRINT_PARAMETER> ')' ';'

<READ_BEGIN> ::= 'read' '(' <READ_END>

<READ_END> ::= <DEC_OBJECT_ATTRIBUTE_ACCESS> ')' ';'

<PRINT_PARAMETER> ::= <DEC_OBJECT_ATTRIBUTE_ACCESS>

| <CAC>

| <NRO>

!-----

! Bloco acesso a atributos e métodos de objetos - Atualizado em 18/09/2023 às 19:39

<MULTIPLE_OBJECT_ATTRIBUTE_ACCESS> ::= <DEC_VAR>

<END_OBJECT_ATTRIBUTE_ACCESS>

<END_OBJECT_ATTRIBUTE_ACCESS> ::= ' '

<MULTIPLE_OBJECT_ATTRIBUTE_ACCESS>

|

<OBJECT_METHOD_OR_OBJECT_ACCESS> ::=

<OBJECT_METHOD_OR_OBJECT_ACCESS_OR_PART>

<OPTIONAL_OBJECT_METHOD_ACCESS> ::= <OBJECT_METHOD_ACCESS_END>

|

<IDE_OR_CONSTRUCTOR> ::= 'constructor'

| IDE

<OBJECT_METHOD_ACCESS_END> ::= '->' <IDE_OR_CONSTRUCTOR> '('

<PARAMETERS> ')'

!-----

! Bloco operadores aritméticos - Atualizado em 16/09/2023 às 22:49

<SIMPLE_OR_DOUBLE_ARITHMETIC_EXPRESSION> ::= <END_EXPRESSION>

| ART_DOUBLE

<END_EXPRESSION_OPTIONAL> ::= <END_EXPRESSION>

|

<SIMPLE_EXPRESSION> ::= <PART> <END_EXPRESSION>

| <PARENTHESIS_EXPRESSION>

<PARENTHESIS_EXPRESSION> ::= '(' <SIMPLE_EXPRESSION> ')'
<END_EXPRESSION_OPTIONAL>

<END_EXPRESSION> ::= ART <PART_LOOP>

<PART_LOOP> ::= <PART> <END_EXPRESSION_OPTIONAL>
| <PARENTHESIS_EXPRESSION>

<PART> ::= <NRO>
| <OBJECT_METHOD_OR_OBJECT_ACCESS_OR_PART>

!-----

! Bloco operadores relacionais - Atualizado em 16/09/2023 às 23:09

<RELATIONAL_EXPRESSION> ::= <RELATIONAL_EXPRESSION_VALUE> REL
<RELATIONAL_EXPRESSION_VALUE>

<RELATIONAL_EXPRESSION_VALUE> ::= <NRO>
| <OBJECT_METHOD_OR_OBJECT_ACCESS>
| <CAC>

!-----

! Bloco operadores relacionais - Atualizado em 16/09/2023 às 22:57

<LOGICAL_EXPRESSION> ::= <LOGICAL_EXPRESSION_BEGIN>
<LOGICAL_EXPRESSION_END>

<LOGICAL_EXPRESSION_BEGIN> ::= '!' <LOGICAL_EXPRESSION_BEGIN>
| '(' <LOGICAL_EXPRESSION> ')'
| <LOGICAL_EXPRESSION_VALUE>

<LOGICAL_EXPRESSION_END> ::= LOG <LOGICAL_EXPRESSION_BEGIN>
<LOGICAL_EXPRESSION_END>
|

<LOG_REL_OPTIONAL> ::= REL <RELATIONAL_EXPRESSION_VALUE>
|

<LOGICAL_EXPRESSION_VALUE> ::= Bool
| <OBJECT_METHOD_OR_OBJECT_ACCESS>
<LOG_REL_OPTIONAL>

!-----

! Bloco for - Atualizado em 18/09/2023 às 10:40

<FOR_BLOCK> ::= <BEGIN_FOR> <FOR_INCREMENT> <END_FOR>

<ASSIGNMENT> ::= '=' <VALUE>
| ART_DOUBLE

<FOR_INCREMENT> ::= <DEC_OBJECT_ATTRIBUTE_ACCESS> <ASSIGNMENT>

<BEGIN_FOR> ::= 'for' '(' <OBJECT_ACCESS_OR_ASSIGNMENT> ';' <CONDITIONAL_EXPRESSION> ';' <END_FOR> ::= ')' '{' <COMMANDS> '}'

<CONDITIONAL_EXPRESSION> ::= <RELATIONAL_EXPRESSION> | '(' <RELATIONAL_EXPRESSION> ')' !-----

! Bloco de métodos - Atualizado em 18/09/2023 às 11:55

<METHODS_BLOCK> ::= 'methods' '{' <METHODS> '}'

<METHODS> ::= <METHOD> <METHODS> |

<METHOD> ::= <TYPES> IDE '(' <DEC_PARAMETERS>

<TYPES> ::= 'void' | <TYPE_VARIABLES>

<TYPE_VARIABLES> ::= <TYPE> | IDE

! ----- Parameter Declaration -----

<DEC_PARAMETERS> ::= <VARIABLE_PARAM> <MULT_DEC_PARAMETERS> | <OBJECT_PARAM> <MULT_DEC_PARAMETERS> | <END_DEC_PARAMETERS>

<MULT_DEC_PARAMETERS> ::= ',' <TYPE_VARIABLES> IDE <MULT_DEC_PARAMETERS> | <END_DEC_PARAMETERS>

<END_DEC_PARAMETERS> ::= ')' '{' <METHOD_BODY>

! ----- Method Body -----

<METHOD_BODY> ::= <VARIABLES_BLOCK> <OBJECTS_BLOCK> <COMMANDS_METHOD_BODY>

! ----- Commands -----

<COMMANDS_METHOD_BODY> ::= <COMMANDS> 'return' <RETURN> ';' '}'

<COMMANDS> ::= <COMMAND> <COMMANDS> |

<COMMAND> ::= <PRINT_BEGIN> | <READ_BEGIN> | <OBJECT_ACCESS_OR_ASSIGNMENT> ';' | <IF> | <FOR_BLOCK>

<RETURN> ::= <VALUE>

|

!-----

! The grammar starts below

<PROGRAM> ::= <CONSTS_BLOCK> <VARIABLES_BLOCK> <CLASS_BLOCK>